

PHYO THANT

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EDUCATION

University of California, San Diego

B.S. in Computer Science (3.6 / 4.0 GPA)

Expected Jun. 2026

- Relevant Coursework: Machine Learning, Software Engineering, Operating Systems, Computer Security, Computer Networks

University of Illinois Urbana-Champaign

Master of Computer Science

Expected Jun. 2028

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL, PHP, JavaScript (HTML/CSS), TypeScript, x86 Assembly

Frameworks & Libraries: FastAPI, PyTest, PyTorch, Scikit-learn, OpenCL, React, Next.js, Node.js, Express.js, MongoDB

Tools: GitHub Actions, Docker, Linux, PostgreSQL, Sanity CMS, Apache, Wireshark, Claude Code

EXPERIENCE

Full-Stack Developer Intern

Think Round Inc.

Jan. 2026 - Present

Remote

- Designed Sanity Studio CMS with structured content schemas, enabling non-technical staff to independently publish and manage pages, eliminating developer content requests
- Migrated high-resolution art archive pages from Squarespace to Next.js + Sanity, leveraging Sanity's image pipeline (AVIF, WebP, lazy loading) to improve page load time from 3s to under 1s
- Leading Vercel deployment and resolved build errors, dependency deprecations, and performance regressions through systematic audits, improving site reliability and content workflow efficiency

Distributed Robotics Engineer Intern

Caltrans

Aug. 2025 - Dec. 2025

San Diego, CA

- Engineered IEEE-funded custom 802.11s mesh radio module with WPA3 encryption, enabling ~100Mbps bandwidth real-time data streaming with <50ms latency
- Enabled real-time remote monitoring of robot telemetry by implementing a low-latency UDP control protocol over the mesh network for sensor streaming, and built a FastAPI + WebSocket dashboard for operators to visualize
- Designed an architecture to auto-connect network topology between Raspberry Pi and LiDAR scanner, eliminating manual setup to begin field inspections immediately on boot-up

Applied ML Undergraduate Researcher

UC San Diego - SEELab

Jul. 2025 - Sep. 2025

San Diego, CA

- Built data pipeline (Python, pandas, NumPy) processing 100k+ multimodal sensor data from LiDAR, camera, and IMU sources, linking agent static attributes with dynamic sensor data across timestamps for trajectory prediction models
- Redesignated data schema from scene-centric to object-centric with rich annotation, reducing model query complexity from $O(\text{scenes} \times \text{cameras})$ to $O(\text{objects})$ and enabling prediction window scaling from 2s to 3s+

PROJECTS

Help Myanmar Rise (Hackathon) | React, MongoDB, Express.js, Node.js, Auth0

- Built a full-stack MERN earthquake relief platform for Myanmar, enabling secure submissions of user reports with Auth0
- Designed interactive Mapbox-powered geospatial visualization with Google Geocoding API integration, improving coordination accuracy by enabling real-time tracking of 1000+ affected regions through dynamic marker clustering
- Automated daily news aggregation using node-cron scheduled jobs and GNews API, reducing redundant external API calls by implementing 24-hour cache expiration logic in Express.js backend

SQL Database Engine | C++, CMake, GoogleTest

- Developed a lightweight database engine in C++ by implementing B+ tree indexing, binary file storage, and a SQL-like query pipeline, resulting in $O(\log n)$ search performance for data lookups and range queries
- Built a custom query parser by applying the shunting-yard algorithm to convert expressions into Reverse Polish Notation (RPN), enabling efficient, stack-based query evaluation
- Designed binary storage engine persisting records to disk in fixed-size format, enabling tables to survive program restarts

AWARDS & CERTIFICATES

MATE ROV World Championship – 13th place (of 79 teams), NASA Robotics Alliance Project

2024

Olsen Tutor Scholarship – Pasadena City College, awarded for excellence in Computer Science tutoring

2024